

Section 2.3 Calculating Limits

$\lim_{x \rightarrow a} f(x)$ try to find $f(a)$

If you get division by zero,
use Algebra!

$$\text{Ex ① } \lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} = \lim_{x \rightarrow 4} \frac{\cancel{(x-4)}(x+4)}{\cancel{x-4}} = 4+4 = \boxed{8}$$

$$\text{Ex ② } \lim_{x \rightarrow 3} \frac{x-3}{x^2-9} = \lim_{x \rightarrow 3} \frac{\cancel{x-3}}{(\cancel{x-3})(x+3)} = \frac{1}{3+3} = \boxed{\frac{1}{6}}$$

$$\begin{aligned} \text{Ex ③ } \lim_{x \rightarrow 100} \frac{(\sqrt{x} - 10)(\sqrt{x} + 10)}{x - 100} &= \lim_{x \rightarrow 100} \frac{\cancel{x-100}^1}{(\cancel{x-100})(\sqrt{x} + 10)} \\ &= \frac{1}{\sqrt{100} + 10} = \boxed{\frac{1}{20}} \end{aligned}$$

$$\underline{\text{Ex (4)}} \quad \lim_{h \rightarrow 0} \frac{(5+h)^2 - 25}{h} = \lim_{h \rightarrow 0} \frac{\cancel{25} + 10h + h^2 - \cancel{25}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(10+h)}{\cancel{h}} = 10 + 0 = \boxed{10}$$

$$\underline{\text{Ex (5)}} \quad \lim_{x \rightarrow -7} \frac{\left(\frac{1}{7} + \frac{1}{x}\right) \cdot 7x}{(7+x) \cdot 7x} = \lim_{x \rightarrow -7} \frac{\cancel{x} + 7}{(\cancel{7} + x) \cdot 7x} = \frac{1}{7(-7)} = \boxed{\frac{1}{-49}}$$

$$\underline{\text{Ex (6)}} \quad \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x^3}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(3x^2 + 3x \cdot h + h^2)}{\cancel{h}} = 3x^2 + 0 + 0$$

$$\boxed{= 3x^2}$$

$$\text{Ex } \textcircled{7} \quad \lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2}$$

$$A^3 + B^3 = (A+B)(A^2 - AB + B^2)$$

$$= \lim_{x \rightarrow -2} \frac{\cancel{(x+2)}(x^2 - 2x + 4)}{\cancel{x+2}} = (-2)^2 - 2(-2) + 4$$

$$= 4 + 4 + 4$$

$$\boxed{= 12}$$

$$\text{Ex } \textcircled{8} \quad \lim_{h \rightarrow 0} \frac{3}{h\sqrt{9+h}} - \frac{1}{h\sqrt{9+h}}$$

$$\lim_{h \rightarrow 0} \frac{(3 - \sqrt{9+h})(3 + \sqrt{9+h})}{(h \cdot \sqrt{9+h})(3 + \sqrt{9+h})}$$

$$\lim_{h \rightarrow 0} \frac{9 - (9+h)}{h \cdot \sqrt{9+h} (3 + \sqrt{9+h})}$$

$$\lim_{h \rightarrow 0} \frac{-\cancel{h}1}{\cancel{h}\sqrt{9+h}(3+\sqrt{9+h})} = \frac{-1}{\sqrt{9+0}(3+\sqrt{9+0})} = \boxed{\frac{-1}{18}}$$